For those already familiar with SDEM

Introduction to Improvements in SDEM® 2.0

Japan Global Gateway

[2022.09.01]

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This training course is intended for those who selected "Introduction to Improvements in SDEM 2.0," for those already familiar with SDEM, in the SDEM 2.0 Introductory Training.

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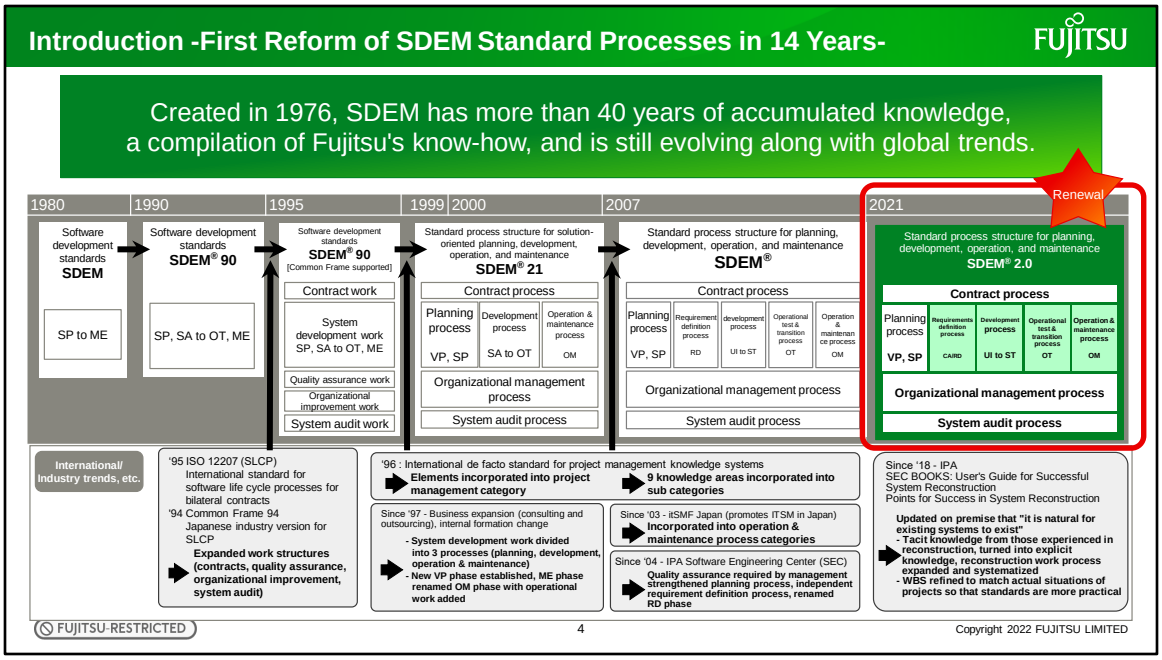

In 2021, SDEM was reformed for the first time in 14 years into SDEM 2.0.
Created in 1976, SDEM has more than 40 years of accumulated knowledge,
a compilation of Fujitsu's know-how, and is still evolving along with global trends.
This training course introduces the improvements made in SDEM 2.0.

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1. Purpose of SDEM 2.0 Renewal

What you will understand by the end of this chapter

- Background to SDEM renewal
- SDEM usage scenarios

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Chapter 1 explains the purpose of SDEM renewal.

By the end of this chapter, you will understand the background to SDEM renewal and SDEM usage scenarios.

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1.1 Purpose of SDEM 2.0 Renewal

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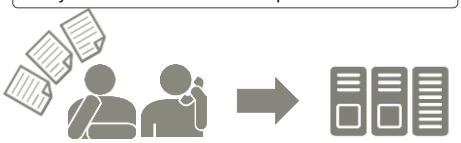
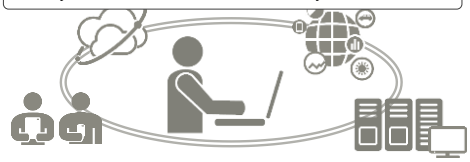
Update that assumes it is natural for existing systems to exist

Old SDEM	Current SI projects
SDEM was established in the early years of core system development, with **the primary theme of making new systems.**	**It is natural for** various **existing systems to exist**. (In new system construction, development cannot proceed without linkage to an existing system.)
Systematization of form development/manual work	System construction in tandem with variety of information





- Tacit knowledge from those with experience in reconstruction projects is turned into explicit knowledge.
- All categories are reformed to be **practical** in line with the reality of current system development.

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The update into SDEM 2.0 assumed that it is natural for existing systems to exist.

The old SDEM was established in the early years of core system development, with the primary theme of making new systems, such as to systematize manual work and the development of forms.
On the other hand, in current SI projects, it is natural for various existing systems to exist around them.
In new system construction, development cannot proceed without linkage to an existing system.

So tacit knowledge from those with experience in reconstruction projects has been turned into explicit knowledge to strengthen processes.
To make all categories practical in line with the reality of current system development, category definitions were reviewed, and purpose and work elements and viewpoints were added to the WBS.

Based on the accumulated know-how in the field, a considerable amount of knowledge to prevent trouble has been added.

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1.2 SDEM 2.0 Usage Scenarios - Can be Used for Both New Construction and Reconstruction -

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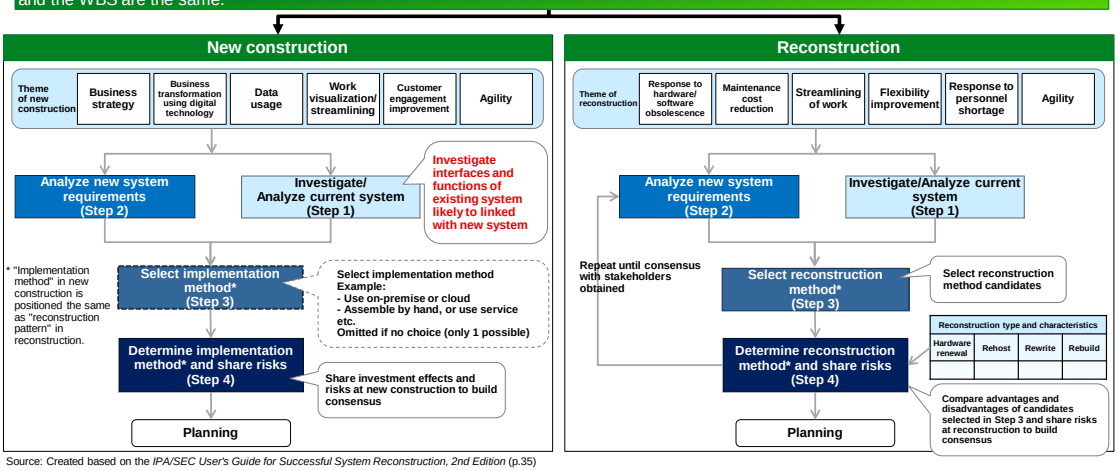
In a world where it is natural for existing systems to exist, investigation of existing systems is necessary for both new construction and reconstruction.
Though the viewpoints differ with the SDEM usage scenario (new construction or reconstruction), the approach to planning for system construction and the WBS are the same.



Source: Created based on the *IPA/SEC User's Guide for Successful System Reconstruction, 2nd Edition* (p.35)

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As mentioned in the purpose, in a world where it is natural for existing systems to exist, investigation of existing systems is necessary for both new construction and reconstruction.
Of course, for reconstruction, the target existing system is investigated.
For new construction, the interfaces and functions of the existing system likely to be linked is investigated in order to utilize a variety of data.

Current analysis in the RD phase in the old version has been systematized and improved as a phase in SDEM 2.0 so that it can be used for both new construction and reconstruction.

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2. Key Points of SDEM 2.0 Renewal

What you will understand by the end of this chapter

- Three key points of SDEM renewal

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Chapter 2 explains key points of SDEM 2.0 renewal.

By the end of this chapter, you will understand three key points of SDEM renewal.

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2.1 Key Points of SDEM 2.0 Renewal

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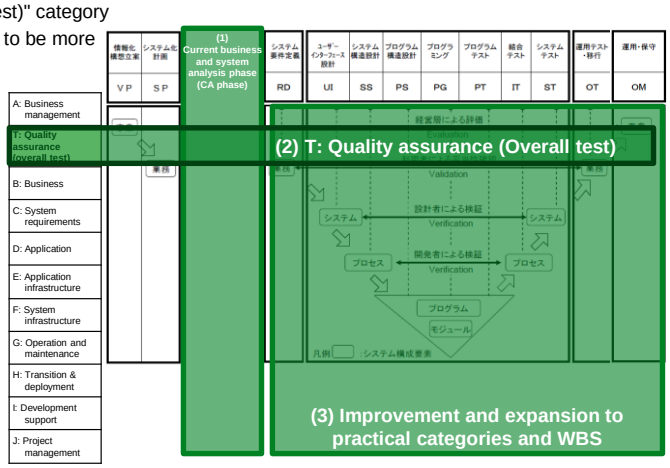
- **Key points of SDEM 2.0 renewal**
  - (1) Added the current business and system analysis phase (CA phase)
  - (2) Added the "T: Quality assurance (Overall test)" category
  - (3) Improved existing categories and the WBS to be more practically.
  - Review of the definitions and positioning of the "B: Business" and "C: System requirements" categories, and addition of sub categories
  - Review of the positioning of the "G: Operation and maintenance" category
  - Refinement and expansion of the "H: Transition & deployment" category and WBS

  etc.

Incorporating the WBS of the *SDEM Practical Standards [Reconstruction Development]*, which was enhanced by new SI quality assurance system initiatives



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The SDEM 2.0 renewal has three significant points:
1. Added the current business and system analysis phase (CA phase)
2. Added the "T: Quality assurance (overall test)" category
3. Improved existing categories and the WBS to be more practical

The third improvement includes, for example, a review of the definitions and positioning of the "B: Business" and "C: System requirements" categories, and the addition of sub categories to them, a review of the positioning of the "G: Operation and maintenance" category, and the refinement and expansion of the "H: Transition & deployment" category and WBS.

SDEM 2.0 incorporates the WBS of the *SDEM Practical Standards [Reconstruction Development]* which was enhanced by new SI quality assurance system initiatives, into SDEM, and has more enhanced additions and reforms.
Chapter 3 introduces the changes specifically in detail.

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2.2 Extended WBS Across Categories

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"Z" indicates work requiring consideration from an overall perspective across categories.

In system development, **some work needs consideration from an overall perspective across SDEM categories**.

"Z" is used in the IDs of items requiring consideration across categories.

Example 1:

Z1CA200 Confirmation of reconstruction requests and tentative definition of reconstruction pattern from requests perspective

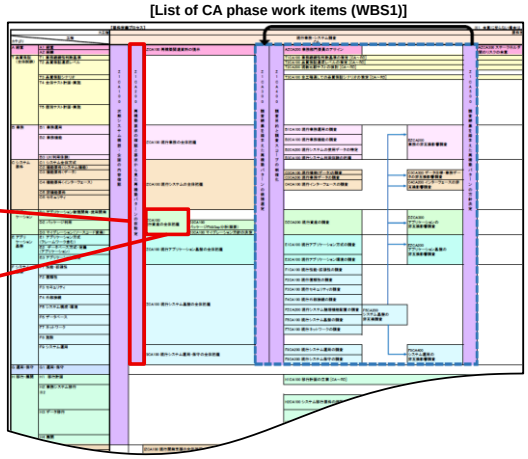
Requiring consideration from an overall perspective across all categories

Example 2:

DZCA100 Overall understanding of current assets

Requiring consideration across sub categories in Category D

[List of CA phase work items (WBS1)]



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SDEM 2.0 features an extended WBS across categories, unlike in the conventional SDEM.

In system development, some work needs consideration from an overall perspective across SDEM categories.

"Z" is used in the WBS-ID of items requiring consideration across categories.

For example, "Z1CA200" is defined for "Confirmation of reconstruction requests and tentative definition of reconstruction pattern from requests perspective" because it requires consideration from an overall system perspective across all categories.
"DZCA100" is defined for "Overall understanding of current assets" because it requires consideration across the entire the "D: Application" category.

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2.3 Definition of Relationships Between WBS Categories

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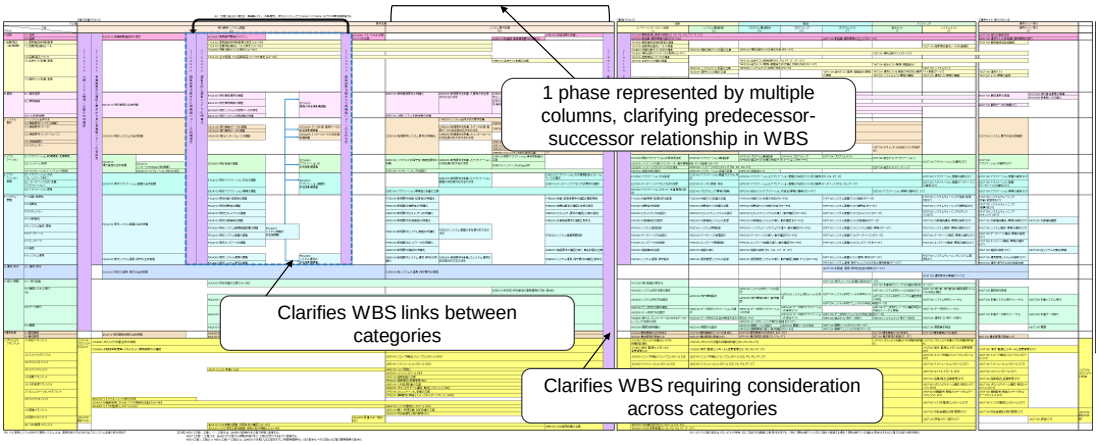
The order and relationship between categories are clearer compared to the old SDEM handbook.



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Another feature of SDEM is the clearer order and relationship between categories compared to the old SDEM handbook.

Especially in the upstream phases such as CA, RD, and UI, a single phase is represented by multiple columns, clarifying the predecessor-successor relationship in the WBS.
Some predecessor-successor relationships are indicated by arrows to clarify WBS links between categories.
A WBS requiring consideration across categories is shown as a vertically long WBS that extends across categories with IDs beginning with "Z" so it is easier to distinguish.

Pay attention to the context and links of the WBS during use.

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3. Changes in SDEM 2.0

What you will understand by the end of this chapter

- Additions and changes related to phases
- Additions and changes related to categories and WBS
- Operation & maintenance process changes

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Chapter 3 explains specific changes made for improvements in SDEM 2.0.

By the end of this chapter, you will understand additions and changes related to phases, additions and changes related to categories and the WBS, and operation & maintenance phase changes.

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3.1 Changes Related to Phases

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First is an explanation of changes related to phases.

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3.1.1 Additions and Changes Related to Phases

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CA phase
- **The WBS focused on current analysis is newly added as "Current business and system analysis" (CA phase).**

RD phase
- **With the addition of the CA phase, the positioning has been changed to determine requirements in line with both the reconstruction requirements and new requirements.**

- Phases in each process

Newly added

Changed

Time axis

Process	Planning process		Requirement definition process		Development process							Operational test & transition process	Operation & maintenance process
Main phase	Planning		Requirement definition		Design		Implementation			Test		Operational test and transition	Operation and maintenance
Phase	IT vision planning	System planning	Current business and system analysis	System requirement definition	User interface design	System structure design	Program structure design	Programming	Program test	Integration Test	System test	Operational test and transition	Operation and maintenance
Abbreviation	VP	SP	CA	RD	UI	SS	PS	PG	PT	IT	ST	OT	OM

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In old version of SDEM, current analysis was one WBS in the RD phase. Current analysis is newly systematized as the CA phase, and the WBS of current business and system analysis has been added. With the addition of the CA phase, the WBS has been expanded to define and determine the requirement definition requirements of the RD phase in line with both the reconstruction requirements and new requirements.

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3.1.2 Addition of CA Phase - Purpose -

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Phase for evaluating if development can actually begin (feasibility and assumed risks)

- Purpose

1. **Determine if the requirement definition or order receipt phase can begin and collect missing information.**
2. **Evaluate feasibility, determine the system construction method, and align awareness with the customer. (Eliminate any discrepancies in awareness.)**
3. **Share possible risks with stakeholders (such as the project owner of the customer) to agree on considerations to hedge risks.**

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This page explains the added CA phase.
The CA phase is the phase for evaluating if development can actually begin, that is, evaluating feasibility and assumed risks.
The CA phase has three purposes.

1. Determine if the requirement definition or order receipt phase can begin during or after development and collect missing information.
2. Based on investigation results, evaluate feasibility, determine the system construction method, and align awareness with the customer to eliminate any discrepancies in awareness.
3. Share possible risks with stakeholders such as the project owner of the customer to agree on considerations to hedge risks.

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3.1.2 Addition of CA Phase - Work Outline -

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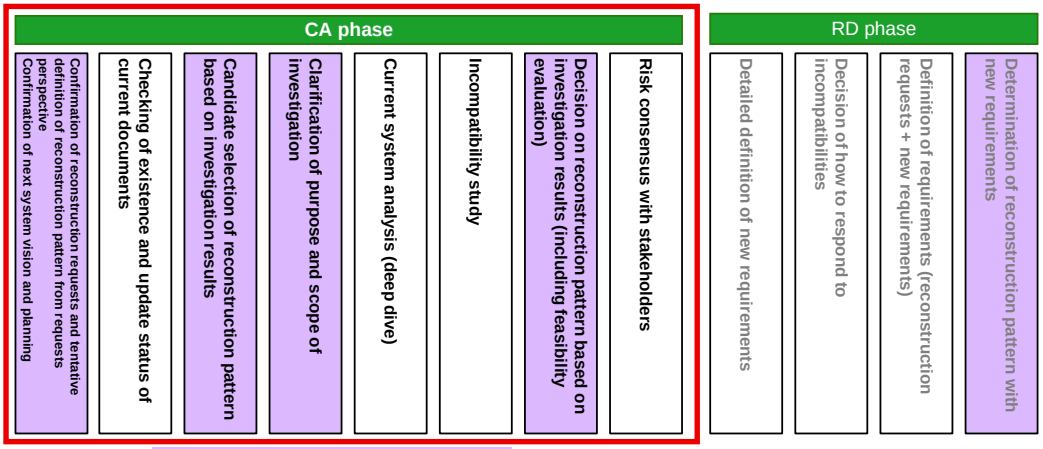
Carry out a current analysis, determine the development methods, and agree on construction risks with stakeholders



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* A category in purple (Z category) is an extended WBS across categories.

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In the CA phase, carry out a current analysis, determine the development methods, and agree on construction risks with stakeholders.

The workflow in the CA phase starts with confirmation of the next system vision and planning from an RFP and tentative definition of the reconstruction pattern from a requests perspective.
Next is the checking of the existence and update status of current documents.
After candidate selection of the reconstruction pattern based on investigation results, the purpose and scope of investigation is clarified from the selected reconstruction pattern.
Based on the clarified purpose and scope of investigation, a current system analysis is performed as a deep dive. Then, an incompatibility study is conducted, especially in modernization using current assets.
Lastly, a decision is made on the reconstruction pattern based on investigation results, which include a feasibility evaluation, followed by a risk consensus with stakeholders.

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3.1.3 Expansion of RD Phase

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Added a WBS for detailed definition of new requirements and determination of requirements meeting requests for reconstruction, and clarified the WBS for considering and deciding measures to respond to incompatibilities

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* Categories in purple (Z category) are WBS extending over all categories

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This page explains the expanded RD phase.
With the addition of the CA phase, the WBS for requirement definition in the RD phase has been expanded for detailed definition of new requirements and the determination of requirements as a new system meeting requests for reconstruction.
Also, a WBS has been added for decisions on how to respond to incompatibilities investigated in the CA phase.

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3.1.4 V-shaped Model for Quality Assurance in System Development

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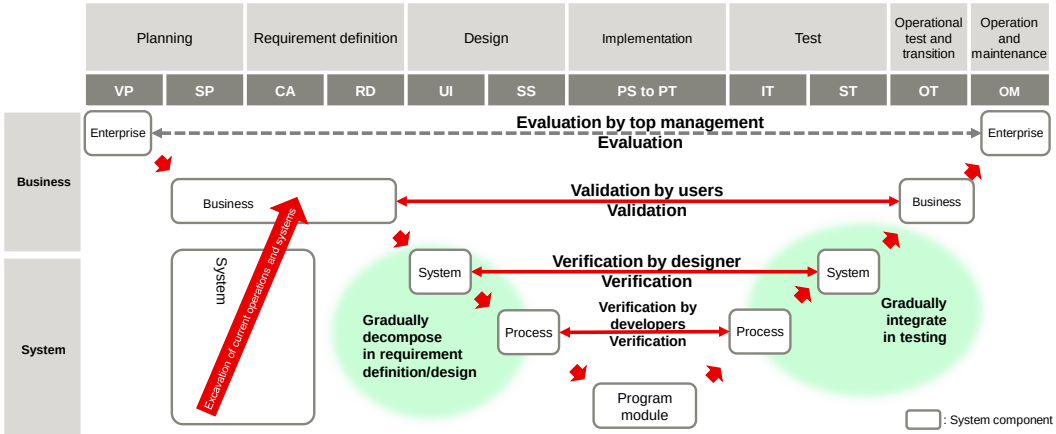
- Entire development process
  - Provide quality assurance by gradually decomposing system components, and gradually integrating them in testing
  - Match the system components at the requirement definition/design stage and the test stage, and verify and confirm quality with the V-shape (V-shaped model for quality assurance)



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In the improvements this time, the concepts of gradual refinement in requirement definition and design and gradual integration in the corresponding test phase have not changed. That is, the V-shaped model for quality assurance in system development is unchanged. To understand and comprehend the concepts, excavation of current operations and systems was added as a precondition.

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3.2 Changes Related to Categories

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This next section explains changes related to categories.

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3.2.1 Additions and Changes Related to Categories

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- **Changes in categories from old SDEM standard processes**

Requirement definition/Development/Operational test and transition process categories

Before changes (old SDEM)

A: Business management	A1: Business management, A2: Organization
B: Business	B1: Business operation, B2: Business function
C: System requirements	C1: Functional requirements (System functions) C2: Functional requirements (Data) C3: Functional requirements (Interface) C4: Non-functional requirements (Category numbers C1 to C4 incremented)
D: Application	D1: Application D2: Package D3: Migration
E: Application infrastructure	E1: Application architectures (including frameworks) E2: Application environment (Category number E2 changed to E3)
F: System infrastructure	F1: Performance & Scalability to F8: Facility
G: Operation and maintenance	G1: Operation and maintenance
H: Transition & deployment	H1: Transition H2: Deployment (Category number H2 changed to H4)
I: Development support	I1: Development standard I2: Configuration management
J: Project management	J1: Integration management to J9: Procurement management

Quality assurance category added
UX (usage experience) category added
Overall system architecture, security categories added
Sub category names changed
Database architectures and definitions category added
System operation category added
Transition categories subdivided
Contract management, OSS application management categories added

After changes (SDEM 2.0)

[Legend] **Red = added**, **Blue= changed**, **Green = definition changed**

A: Business management	A1: Business management, A2: Organization
T: Quality assurance (Overall test)	**T1: Business operation continuity judgment criteria** **T2: Quality assurance achievement level** **T3: Quality assurance scenario** **T4: Overall test plan and execution** **T5: Individual test plan**
B: Business	B1: Business operation, B2 Business function **B3: UX (Usage experience)**
C: System requirements	**C1: Overall system architecture** **C2:** Functional requirements (System functions) **C3:** Functional requirements (Data) **C4:** Functional requirements (Interface) **C5:** Non-functional requirements **C6: Security**
D: Application	D1: **Application - New development and diversion development** D2: **Package usage** D3: **Migration (Source code conversion)**
E: Application infrastructure	E1: Application architectures (including frameworks) **E2: Database architectures and definitions (Applications)** **E3:** Application environment
F: System infrastructure	F1: Performance & scalability to F8: Facility **F9: System operation**
G: Operation and maintenance	G1: Operation & maintenance
H: Transition & deployment	**H1: Transition plan, H2: Business system transition, H3: Data transition, H4: Deployment**
I: Development support	I1: Development standard I2: Configuration management
J: Project management	J1: Integration management to J9: Procurement management **JA: Contract management** **JB: OSS application management**

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The figure shows all the category-related additions and changes from the old version. The tables show a comparison of SDEM before and after the changes.
Here is an explanation of the additions and changes in order from the top.

- "T: Quality assurance (Overall test)" was added as a category for quality assurance.
- The positioning of "B: Business" and "C: System requirements" was reviewed and definitions were changed.
- The UX (usage experience) category was added to B: Business.
- The "C1: Overall system architecture" and "C6: Security" categories were added to C: System requirements.
- The names of sub categories in the "D: Application" category were changed.
- The E2 category was added to E: Application infrastructure, as a category of database architectures and definitions viewed from applications.
- The definition of "G: Operation and maintenance" was changed and the "F9: System operation" category was added accordingly to F: System infrastructure.
- The "H1: Transition" category in "H: Transition & deployment" was subdivided according to actual project conditions.
- Categories related to the management of contracts with customers and OSS applications were added to J: Project management.

The following pages explain the contents of the changes.

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3.2.2 T: Quality Assurance (Overall Test) - Purpose -

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Systematization of overall management for quality assurance in production

- Purpose

1. Take risk measures at the project startup stage
Predict and detect quality assurance risks due to project/customer characteristics at an early stage of the project, and **enable negotiations and agreement with the customer and execution of risk measures from the time of project planning**.

2. Take early risk measures while the project is in progress
Predict and determine from the risk detection stage which parts of an overall test plan in an early stage would be impacted by risks (progress delay, additional requirements, troubles, etc.) that occur on a daily basis in an ongoing project, thereby **enabling engagement in customer negotiations and risk measures from an early stage**.

- Sub categories

T1: Business operation continuity judgment criteria	Establish business operation continuity judgment criteria. Evaluate business operation continuity as an operational judgment.
T2: Quality assurance achievement level	Formulate quality goals to be achieved by the new system, proceed toward achievement of the goals.
T3: Quality assurance scenario	Formulate a scenario to build quality for production. **This scenario serves as a model for all quality assurance**.
T4: Overall test plan and execution	Based on a quality assurance scenario, develop a test plan for the entire system, and proceed with testing.
T5: Individual test plan	Based on an overall test plan, develop individual test plans (by business, by category, etc.).

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The meaning and purpose of the newly established "T: Quality assurance (Overall test)" category are to understand the processes from planning and test execution to final judgment in order to provide quality assurance in production. Then, project management can proceed. Category T covers quality assurance through testing out of quality assurance through design and testing.

Category T has two purposes.
The first is to take risk measures at the project startup stage.
Predict and detect quality assurance risks due to project/customer characteristics at an early stage of the project, and enable negotiations and agreement with the customer and execution of risk measures from the time of project planning.

The second is to take early risk measures while the project is in progress.
Predict and determine from the risk detection stage which parts of an overall test plan in an early stage would be impacted by risks that occur on a daily basis in an ongoing project, thereby enabling engagement in customer negotiations and risk measures from an early stage.

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3.2.2 Quality Assurance (Overall Test) - Flow From Quality Scenario -

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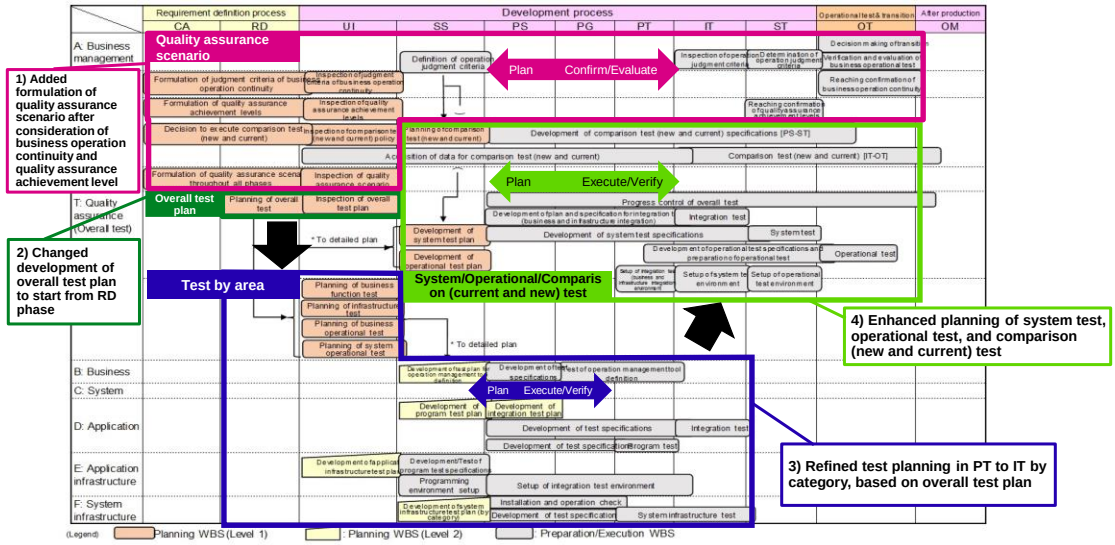

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This page explains the major flow from quality assurance scenario to planning and execution of test including added
Category T and improvement points.
The figure shows excerpts from the WBS related to the test.

1) First, consider business operation continuity and the quality assurance achievement level in the CA and RD phases, and formulate a quality assurance scenario. This WBS is newly added.
2) Next, develop an overall test plan based on the quality assurance scenario. The overall test plan was reviewed to start consideration from the RD phase.
3) Based on the overall test plan, formulate test plans from PT to IT by category area. The WBS for test planning for each area was added.
4) Based on the quality assurance scenario and the overall test plan, integrate tests by area and formulate system, operational, and comparison (new and current) test plans. The planning WBS was enhanced, and the WBS for new and current tests was added.

For your reference, the *WBS Manual* describes points to consider about each test plan.

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3.2.3 B: Business / C: System Requirements

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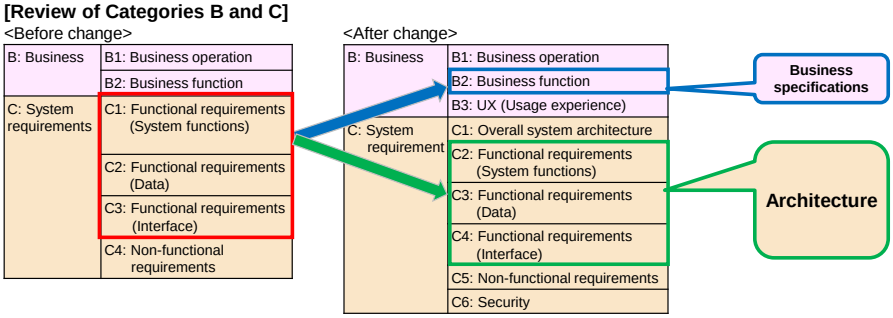
Reviewed the definitions and positioning of "B: Business" and "C: System requirements"

- **Consolidated all perspectives related to the business specifications of customers into "B: Business"**
- **Turned Category C into the specialized role of "architecture"**

[Review of Categories B and C]



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This and subsequent pages explain the changes in Categories B and C.

The definitions and positioning of "B: Business" and "C: System requirements" were reviewed.

Perspectives related to the business specifications of customers were all consolidated into "B: Business." The positioning of "C: System requirements" was clarified to turn this category into the specialized role of "architecture."

Next, I will explain sub categories we added.

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3.2.3 B: Business / C: System Requirements - B3: UX (Usage Experience) -

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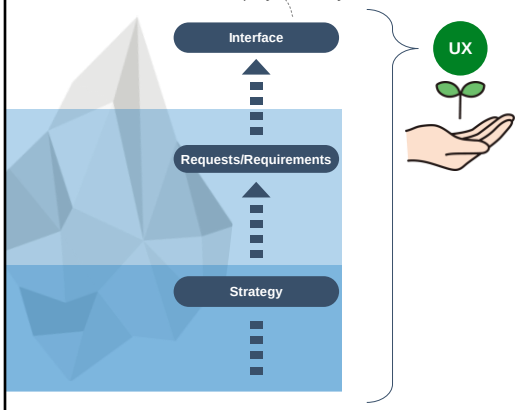
Added a review of UX (user experience) as the "B3: UX (Usage experience)" category

Part of the UX is the screen, the top layer of the system.



Customers' experience value maximized by the UX category

The UX category was added to **maximize experience value** by considering, from the user perspective, how to utilize the system and what will be accomplished.

In light of issues in the field and the **strategy** to maximize the effect of the next system, **requests and requirements (what is required for the system)** were clarified. Based on the strategy, experience value is enhanced by detailed definition of the necessary functions and interface.

Not only user-friendly screens but the **work items and processes necessary for designing and developing the system will be the most valuable to customers**.

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The review process regarding user experience (UX) was newly established in "B: Business" as the "B3: UX (Usage experience)" category.

The UX (usage experience) category considers how to utilize the system and what will be accomplished from the user perspective. Not only user-friendly screens but the work items and processes necessary for designing and developing the system will be the most valuable to customers.

In addition to the requirement definition work performed in the traditional system development, requirements are defined from the user perspective to keep in mind how to utilize the system and what will be accomplished. This will lead to a purpose-oriented proposal and system design.
An approach starting from that experience will lead to solutions to customers' problems, which in turn enhances business competitiveness.

This category has two purposes.

The first is to clarify what is ideal for the next system and what the system should do in line with real requests from the field. Using that image of a system as a basis for discussion, the parties concerned can proceed after agreeing on what the next system should be.

Second is to define in detail the screens that will produce important user experiences during the RD phase and share them with the parties concerned to:
1) make it easy to recall experiences after system operation begins, and
2) allow users to check if the ideal has been realized, which makes it possible to correct the parts likely to become a significant problem.

For details of work related to this category, see the *SDEM Practical Standards: UX Design (ITUMAP)*.

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3.2.3 B: Business / C: System Requirements - C1: Overall System Architecture -

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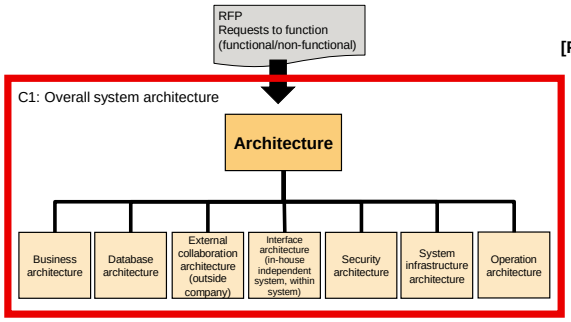
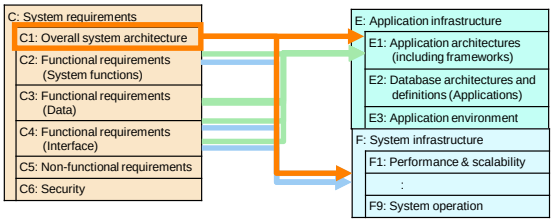
Added the consideration of the overall system architecture and architecture design as "C1: Overall system architecture"

[C1: Overall system architecture structure]



[Flow of consideration from Category C to each category]



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The "C1: Overall system architecture" category was added to "C: System requirements," which considers and designs the overall system architecture.
This category was added because it is important to consider and design the overall system architecture for both new construction and reconstruction.
Insufficient consideration of the architecture will have a significant impact on the entire system, such as the system implementation method, project costs, schedule, quality, and so on.

Requirement definition for elements in the overall system architecture, including the business architecture, database architecture, external collaboration architecture, interface architecture, security architecture, system infrastructure architecture, and operation architecture from RFP functional requests and non-functional requests was systematized.
From "C1: Overall system architecture," the system level architecture, including the OS, middleware, and virtual environment, connects to the system infrastructure, and the application-related architecture connects to the application infrastructure.

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3.2.3 B: Business / C: System Requirements -C6: Security-

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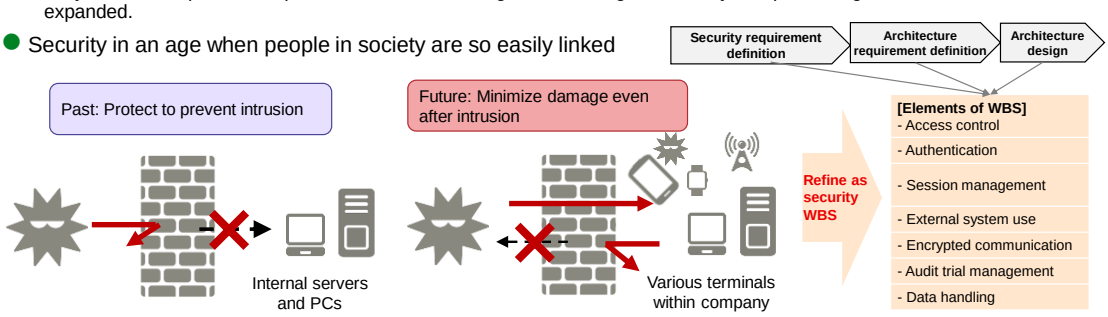
Added the WBS for considering security requirements for the overall system as "C6: Security"

- Background to adding "C6: Security"

  In reconstruction (for Growth) for DX. etc., greater collaboration with outside parties on data utilization will require security measures to be taken.

  In system development, the process aimed at realizing an ideal image of security and preventing work omissions has been expanded.

- Security in an age when people in society are so easily linked



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The WBS for considering security requirements for the overall system and design architecture from requirement definition is newly systematized as "C6: Security."

Category F already has a security category. However, greater collaboration with outside parties on data utilization in growth fields aimed at DX, etc. will required more secure measures to be taken.
Category C6 was added and the WBS was expanded with the aim of realizing an ideal image of security in the overall system and application development, not only in the infrastructure.

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3.2.4 D: Application / E: Application Infrastructure

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Reviewed every WBS in Categories D and E for practical use

- **D: Application**
  - To clarify the work in sub categories, **sub category names were reviewed and any WBS inadequate for practical use was improved**. The following was added, for example:
    - WBS to analyze existing assets (such as programs and definition objects) to the CA phase
    - WBS to clarify implementation of application performance and verification work
    - "Development of program test plan" WBS, "Development of integration test plan" WBS, and WBS for preparation of data for each test
- **E: Application infrastructure**
  - **With the addition of "C1: Overall system architecture," the WBS related to implementation in the application infrastructure was clarified accordingly**.
    - The "Defining list of application infrastructure functions that cannot be realized by frameworks" WBS was added.
  - **The WBS for performance implementation specification/rule definition was expanded**.
    The following was added, for example:
    - "Definition of non-functional specification to be included in applications" WBS and "Performance modeling (business)" WBS
    - "Definition of business application design rules based on the architecture" WBS
  - "E2: Database architectures and definitions (Applications)" is newly systematized.
    - Each WBS from "Design of database access architecture" to "Table implementation and development" was added.

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This page explains the changes in Categories D and E.

The sub category names for "D: Application" were reviewed in order to clarify the work of each category, and any WBS inadequate for practical use was improved.
For example, the following was added:
- WBS to analyze existing assets such as programs and definition objects
- WBS to clarify implementation of application performance and verification work
- "Development of program test plan" WBS, "Development of integration test plan" WBS, and WBS for preparation of data for each test

With the addition of "C1: Overall system architecture," the WBS related to implementation in the application infrastructure was clarified accordingly in "E: Application infrastructure," such as "Defining list of application infrastructure functions that cannot be realized by frameworks."
Also, the WBS for performance implementation specification/rule definition was added.

The "E2: Database architectures and definitions" category was also added, and each WBS from "Design of database access architecture" to "Table implementation and development" was added.
That is because Category D contained database architecture and definition work and role division in Category F was unclear. So work as sub categories of "E: Application infrastructure" was clarified.

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3.2.5 F: System Infrastructure / G: Operation and Maintenance

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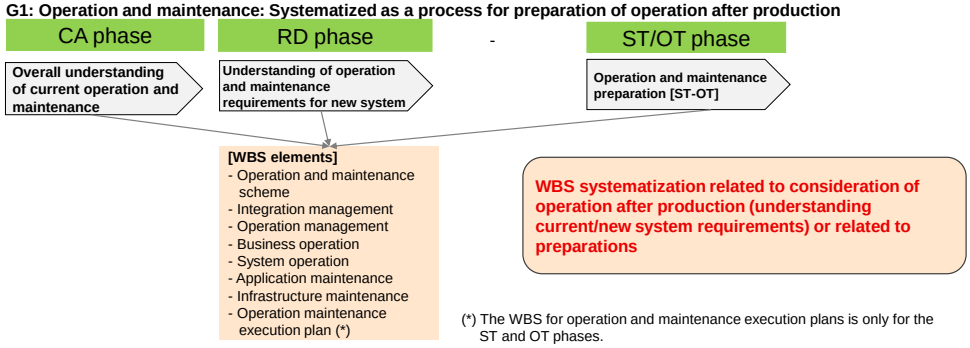
Reviewed the definition and positioning of the "G: Operation and maintenance" category, and added "F9: System operation" to Category F

- **F: System infrastructure**
  - **F9: System operation: Changed the positioning of the WBS related to construction of system operation of "G1: Operation and maintenance" in the old version to system infrastructure**
- **G: Operation and maintenance**
  - **G1: Operation and maintenance: Systematized as a process for preparation of operation after production**



(*) The WBS for operation and maintenance execution plans is only for the ST and OT phases.

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This page explains the changes in Categories F and G.

"G1: Operation and maintenance" in the old version mainly deals with infrastructure construction work for system operation. Preparation work for operation after production was insufficient.
Therefore, "G1: Operation and maintenance" in SDEM 2.0 was reviewed as a preparation process for operation and maintenance after production.
Accordingly, "F9: System operation" in Category F was added, and the WBS related to system operation construction in Category G1 in the old version was moved.

The WBS for understanding current operations and maintenance in the CA phase of "G1: Operation and maintenance," for understanding operations and maintenance requirements for the new system in the RD phase, or for operation and maintenance preparation after production in the ST and OT phases was systematized.

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3.2.6 H: Transition & Deployment

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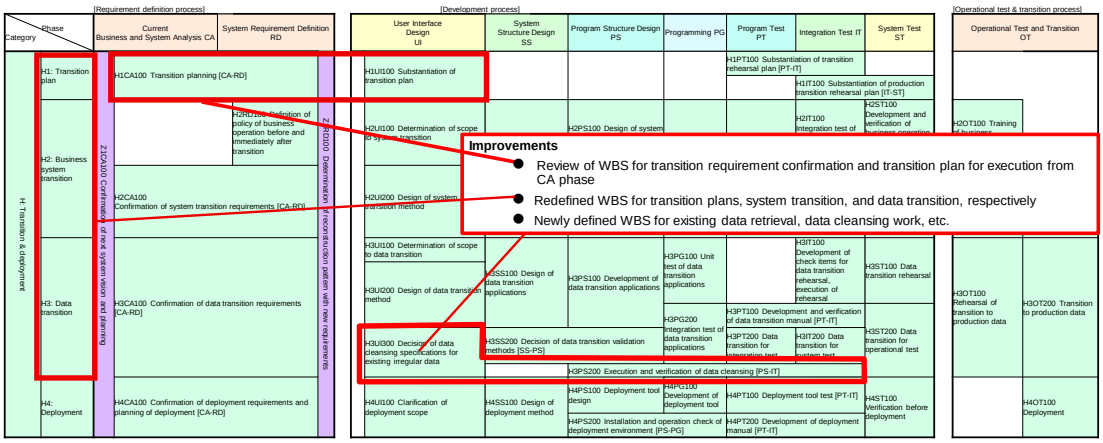
- Changed to start transition planning in an earlier step to determine transition risks at an early stage
- Refined and expanded WBS in line with project implementation



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This page explains the changes in Category H.

Transition planning in "H: Transition & deployment" was positioned from the CA phase in order to determine transition risks at an early stage. Also, each WBS was refined or expanded in line with project implementation.

There are three improvements.
1. The WBS for transition requirement confirmation and transition plan was reviewed for execution from the CA phase.
2. The WBS for the following sub categories was redefined: Transition plan, System transition, and Data transition.
3. The WBS for existing data retrieval, data cleansing work, etc. is newly defined.

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3.2.7 J: Project Management

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Systematization of risk management of contracts and OSS applications

- "JA: Contract management" category added

- **"JB: OSS application management" category added**
  - Investigation of OSS usage and confirmation of terms of use
  - Clarification of scope of responsibility and role division for OSS
  - OSS license compliance plan
  - OSS license compliance control
  - OSS change management

J: Project management	J1: Integration management
	J2: Scope management
	J3: Time management
	J4: Cost management
	J5: Quality management
	J6: Human resource management
	J7: Communications management
	J8: Risk management
	J9: Procurement management
	JA: Contract management
	JB: OSS application management

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This page explains the changes in Category J.

"J: Project management" systematizes the WBS for the management of contracts with customers and OSS applications.

The WBS to define the process and timing of contracts with customers was added the "JA: Contract management" category. This includes proposal and explanation of responsibility sharing and conditions for entrustment to avoid discrepancies in awareness with customers, negotiation to overcome risks, and record of agreement that will serve as normative guidelines.

The WBS for consideration of the correct use of OSS and license compliance control was added the "JB: OSS application management" category.

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3.3 Operation & Maintenance Process Changes

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This conclusion to the section on changes explains the changes in the operation & maintenance process.

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3.3.1 Changes in Operation & Maintenance Process Categories

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ITIL concepts already addressed in the *SDEM Practical Standards [Operation and Maintenance (ITSMOP)]* are incorporated, and category definitions have been reviewed.

- **Operation & maintenance process (OM phase)**

Before change (old SDEM) [Legend] Blue = before change

A: Business management	A1: Business management A2: Organization
B: Business	B1: Business operation B2: Business function
C: Service delivery	C1: Service level management to C6: Information security management
D: Service support	D1: Service desk to D6: Configuration management
E: System operation	E1: System operation plan management to E4: System operation training
F: System maintenance	F1: System maintenance plan management to F6: Maintenance support
G: Project management	G1: Integration management to G9: Procurement management

Incorporating latest ITSMOP

Integrating system operation and maintenance category

After change (SDEM 2.0) [Legend] **Blue = after change**

A: Business management	A1 Business management A2 Organization
B: Business	B1: Business operation B2: Business function
C: IT service management	**C1: Integrated management** **C2: Operational management**
D: System operation and maintenance	**D1: Business operation** **D2: System operation** **D3: Application maintenance** **D4: Infrastructure maintenance**
E: Infrastructure provision	**E1: Infrastructure provision**
G: Project management	G1: Integration management to G9: Procurement management

Note: For an explanation of a WBS for Category C, D, or E of the operation and maintenance (OM) phase, see the *SDEM Practical Standards (Operation and Maintenance: ITSMOP)*.

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ITIL concepts already addressed in the *SDEM Practical Standards [Operation and Maintenance (ITSMOP)]* are in the OM phase of the operation & maintenance process, and category definitions have been reviewed.

SDEM and ITSMOP are now consistent.
For an explanation of a WBS for Category C, D, or E of the operation and maintenance (OM) phase, see the *SDEM Practical Standards (Operation and Maintenance: ITSMOP).*

4. SDEM 2.0 Content Structure

What you will understand by the end of this chapter

- What kind of content is provided with SDEM
- How far the SDEM provision scope extends
- Where to obtain SDEM, and contact information

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Chapter 4 introduces the content structure of SDEM 2.0.

By the end of this chapter, you will understand what kind of content is provided with SDEM, how far the SDEM provision scope extends, where to obtain SDEM, and contact information.

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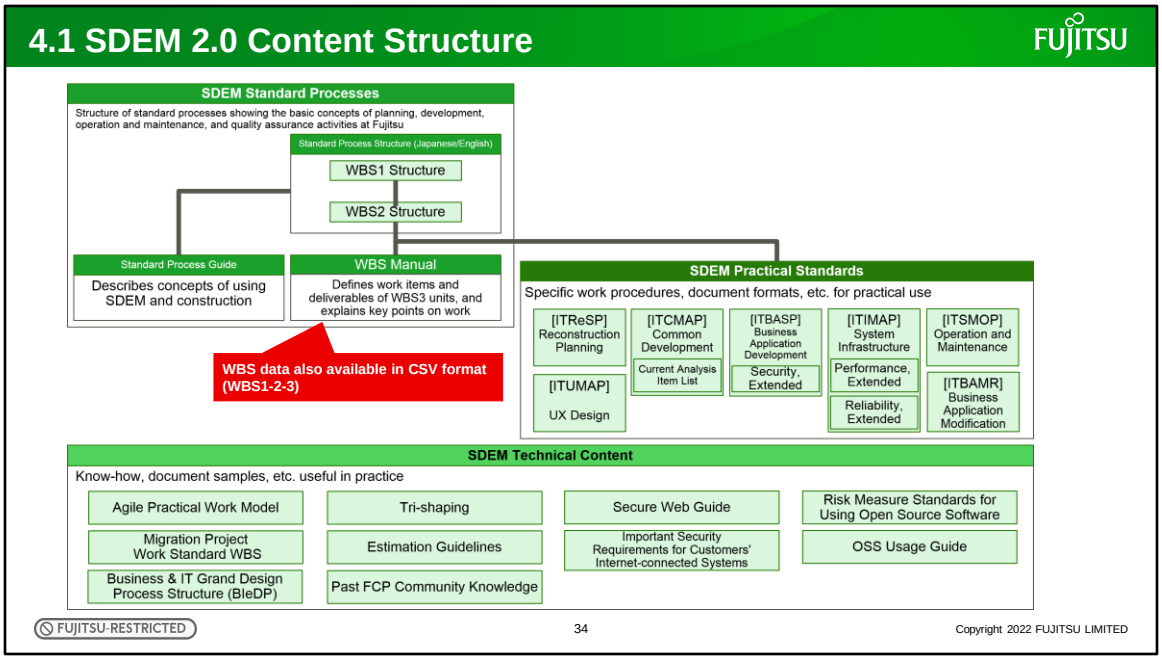

The SDEM content structure broadly consists of three types of content: standard processes, practical standards, and technical content.

The SDEM Standard Processes include the *Standard Process Structure*, *WBS Manual,* and *Standard Process Guide*.
The *Standard Process Structure* shows the WBS1 and WBS2 structures and phase and category definitions.
The *WBS Manual* defines the work items and deliverables of WBS3 units and explains key points about the work.
The *Standard Process Guide* explains SDEM use.

The WBS1-2-3 lists are also available in CSV format for easy customization in a project.

[Reference] Standard Process Guide Excerpt 1) Concept of CA Phase

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Chapter 4 Requirement Definition/Development/Operational Test and Transition Process

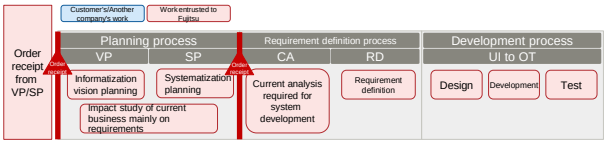
[Point] Concept of CA phase

Customers primarily handle the work in the current business and system analysis phase. Nonetheless,
Fujitsu takes a proactive approach to avoid development risks.

- Clearly define and reach an agreement on role division between the customer and Fujitsu on current analysis work.
- Be sure to obtain the customer's approval and judgment on work performed by Fujitsu.

See this to find out more about the concept of CA phase and current analysis levels.

- Current analysis differences between the VP/SP and CA phases
  - Current analysis in the VP/SP phase mainly investigates **requests (what to change or not change)**.
    - ✓ Current business/system discovery at an outline level
    - ✓ New business/system images incorporating requests
    - ✓ Feasibility verification of requests
    - ✓ Policies during and after the next phase (Reconstruction pattern, quality assurance policy, approximate cost estimation, project plan, etc.)
  - Current analysis in the CA phase is the **current analysis required for system development**. **Be sure to execute it**. Evaluate sufficiency for a current analysis, and investigate missing items that require further investigation.
- Orders received from VP/SP



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Here is supplementary information about the *Standard Process Guide*.

The *Standard Process Description* is a physical book with the old SDEM. Nicknamed the green book, it is now available in PDF format as the *Standard Process Guide* in light of today's work environment.
The written contents range from the basic concepts of SDEM to points to keep in mind in reconstruction.
Each published chapter is available for browsing on the Web. Browse according to the topic you want to know about.

For example, "Chapter 4 Requirement Definition/Development/Operational Test and Transition Process" covers the concept of CA phase.
The chapter describes the concept of current analysis execution at the planning stage or the order receipt time.
See this chapter to find out more about current analysis differences between the VP/SP phase and CA phase and the concept of current analysis when an order is received from the UI phase.

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[Reference] Standard Process Guide Excerpt 2) Appendix 5. SDEM 2.0 Details of Changes

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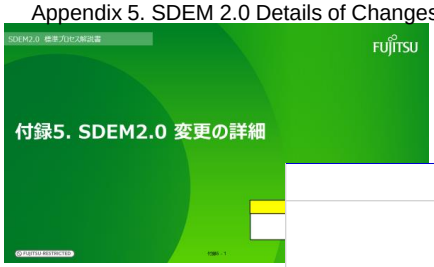
Appendix 5. SDEM 2.0 Details of Changes



SDEM and SDEM 2.0 correspondence list

* The figure below shows the correspondence between SDEM (version 1.0.1a) and SDEM 2.0 (version 1.0).

See the appendix to find out the correspondence between the old SDEM and SDEM 2.0 by WBS.

SDEM－SDEM2.0 対応一覧　WBS対応（要件定義プロセス～運用テスト・移行プロセス）

※変更区分
'追加'：SDEM2.0で追加した作業項目
'削除'：SDEM2.0で削除した作業項目
'移動'：SDEM2.0で大または小カテゴリを移動した作業項目(カテゴリ追加によってカテゴリコードがずれたものは除く)
'変更'：'移動'以外でSDEM2.0でWBS1、WBS2のコードまたは名称を変更した作業項目
'－'　：WBS1、WBS2のコードおよび名称の変更なしの作業項目　※ただし、WBS3の変更はある可能性があります

SDEM				SDEM2.0					
WBS1		WBS2		変更	移動先	WBS1		WBS2	
コード	名称	コード	名称	区分		コード	名称	コード	名称
				追加	－	B1RD300	業務運用要件の具体化	B1RD350	帳票運用の定義
				追加	－	B1RD300	業務運用要件の具体化	B1RD360	外部連携ファイル運用の定義
				追加	－	B1RD300	業務運用要件の具体化	B1RD370	マスタデータ運用の定義
B1UI100	業務運用、新業務データの設計［ＵＩ～ＳＳ］	B1UI110	システム化対象外の業務の定義	削除					
B1UI100	業務運用、新業務データの設計［ＵＩ～ＳＳ］	B1UI120	業務運用の設計	変更	－	B1UI100	業務運用の設計	B1UI110	業務運用の設計
B1UI100	業務運用、新業務データの設計［ＵＩ～ＳＳ］	B1UI130	新業務データの設計［ＵＩ～ＳＳ］	移動	B2UI210				
B1SS100	業務運用テスト方針の決定	B1SS110	業務運用テスト方針の決定	移動	T4SS210				

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This appendix in the guide provides the details of changes between the old SDEM and SDEM 2.0.

See this appendix to find out the details of the correspondence between the old SDEM and SDEM 2.0 by WBS.

For your reference, the history of changes in the latest version is available on the page about SDEM 2.0 standard processes on the IKB website.

4.2 Structure of SDEM Practical Standards

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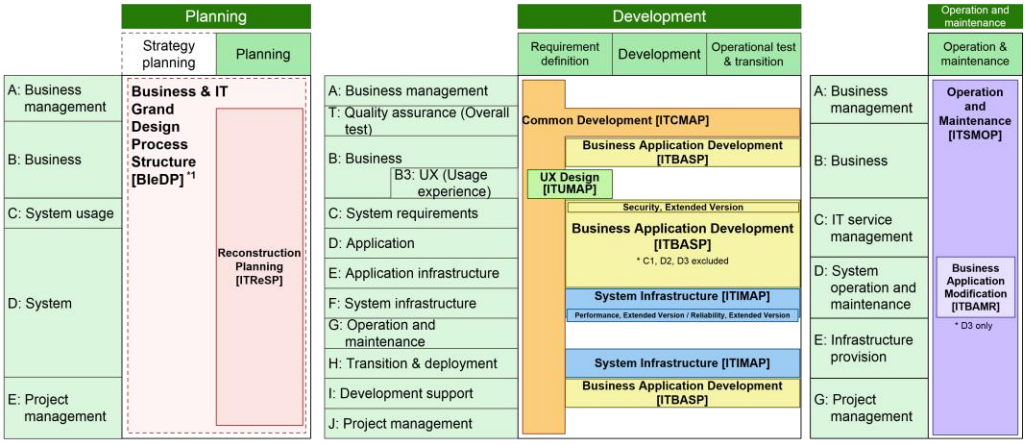
The SDEM Practical Standards provide specific work procedures, document formats, etc. in practice.



*1 The *Business & IT Grand Design Process Structure (BleDP)* is available in the SDEM Technical Content.

*** For the location of document formats and samples, see "ドキュメントフォーマット・サンプル格納場所" (Document format/sample storage location) on the following page:**
https://fujitsu.sharepoint.com/sites/jp-IKB-sdem/SitePages/technology/sdeminformation.aspx

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Next is an introduction to the *SDEM Practical Standards*.

The *SDEM Practical Standards* provide specific work procedures, document formats, etc. in practice.
This figure shows the correspondence between titles in the SDEM Practical Standards and phases and categories in the SDEM Standard Processes.
Select a title according to the phase or category you want to know about.

Every WBS in the *Reconstruction Development, Improved [ITMMAP]* improved by new SI quality assurance system initiatives has been incorporated into the SDEM 2.0 standard processes.
Accordingly, the list of current analysis items and the *Quality Assurance (Overall Test) Guidelines*, which used to be provided with ITMMAP, are now available as the *Common Development [ITCMAP]*.

The subsequent pages introduce the newly added content.

[Reference] Business & IT Grand Design Process Structure [BIeDP*]

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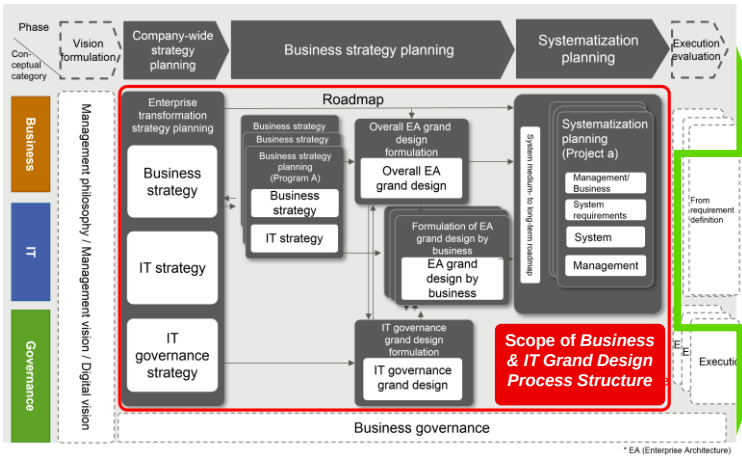
Systematize the work processes and knowledge for the DX vision/planning



* BIeDP:
Business & IT Enterprise Design Process

To requirement definition and development processes in SDEM

The *Business & IT Grand Design Process Structure* is available in the SDEM Technical Content.

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For business transformation utilizing the latest digital technologies, IT human resources familiar with digital technologies need to participate from company-wide strategy planning or business strategy planning, not from system planning or requirement definition.

Customers should play the main role in developing the company-wide strategy and business strategy. However, as a DX partner, we should understand the customers' ideas and provide support as an IT expert.

The *Business & IT Grand Design Process Structure [BIeDP]* systematizes the work processes and knowledge for the DX vision/planning.
Use it to support the customers' DX vision/planning.

[Reference] SDEM Practical Standards: UX Design (ITUMAP)

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- Provides **documents that can readily** describe for **putting into practice** the work in the "B3: UX (Usage experience)" category
  - Sample document: Entry examples and formats for output
  - *WBS Manual Supplementary Materials*: Provides detailed descriptions and know-how in an operation guide

- Please make the most of the **content for learning** about UX design and the **support service** for the projects.
  - Content for learning about UX design: URL: https://fujitsu.sharepoint.com/sites/jp-cx/SitePages/education.aspx
  - Experience design support service: URL: https://fujitsu.sharepoint.com/sites/jp-cx
  - "Experience Design for All" on Yammer: URL: https://qis590.fjqis.css.fujitsu.com/y/l4dd

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The *SDEM Practical Standards [UX Design (ITUMAP)]* provides the documents necessary to understand the work in the "B3: UX (Usage experience)" category.
They include specific work procedures, know-how, and document formats for practical use.

Content related to UX design will continue to be released.
Aside from the ongoing release of ITUMAP documents, training is offered to learn about UX design and UI design.

UX design experts also participate in the projects. Services closely aligned with SDEM are provided to support the implementation of various processes in the "UX (Usage experience)" category and the acquisition of skills.
You can use the skills you acquire in your projects.

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[Reference] ITBASP Security, Extended Version

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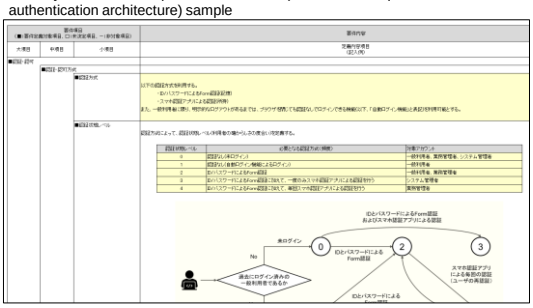
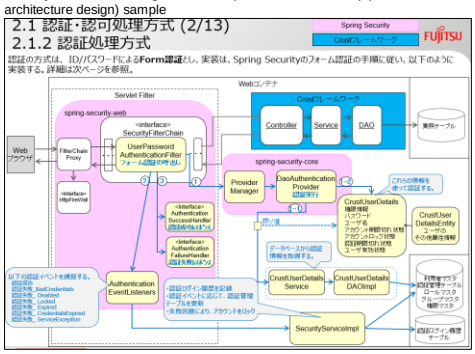
- Provides the know-how necessary for putting into practice "C6: Security" which is improved in SDEM 2.0
- Written in collaboration with the members of the Data & Security Research Laboratory who wrote the *Secure Web Guide*
- Provides document samples and formats that are so practical that even beginners in security can easily understand the key points of requirement definition and design for security

C6RD320t1
Security architecture requirement definition (Definition of requirements for authentication architecture) sample



C6RD420t1
Security architecture design document (common architecture) (Authentication architecture design) sample



Business application development website ◆*ITBASP Security, Expanded Version*
https://fujitsu.sharepoint.com/sites/jp-IKB-sdem/SitePages/technology/practicestandard/itbasp.aspx

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Last is the *ITBASP Security, Extended Version*.
The *ITBASP Security, Extended Version* provides the know-how necessary for putting into practice "C6: Security" which was improved this time. The provided document samples and formats are so practical that even beginners can easily understand the key points of requirement definition and architecture design. Please make the most of them.

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4.3 Scope of SDEM 2.0 Provision

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SDEM is Fujitsu's core technology and important know-how for Fujitsu to outsource SI. Therefore, provision to an external party is based on the premise that an NDA is concluded and Fujitsu is entrusted with SI.

SDEM is copyrighted by Fujitsu, and the scope of provision varies depending on the content.

SDEM Structure		Internal Provision Scope	External Provision Scope	
		Within Company/Group	Business Partner	Customer
SDEM Standard Processes	Standard Process Structure	✓	◆	◆
	WBS Manual	✓	◆	◆
	Standard Process Guide (FUJITSU-CONFIDENTIAL)	✓	◆	◆
	Standard Process Guide (for internal use)	✓	x	x
	WBS1-2-3 work item list (CSV format)	✓	x	x

[Legend]
✓: Can be provided
◆: Can be provided after the conclusion of a non-disclosure agreement (NDA), representing a confidentiality agreement, on the premise that Fujitsu is entrusted with SI
x: Cannot be provided or shown

* For the provision scope of the SDEM Practical Standards, see "利用上の注意事項" (Precautions for Use) on the following page:
https://fujitsu.sharepoint.com/sites/jp-IKB-sdem/SitePages/technology/attention.aspx

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Here is an explanation of the scope of provision of SDEM 2.0.

SDEM is Fujitsu's core technology and important know-how for Fujitsu to outsource SI.
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技術情報
- 利用上の注意事項
- SDEM総合案内
- SDEM標準プロセス
- SDEM実践標準
- ・SDEM実践標準体系
- ・再構築企画編
- ・開発共通編
- ・UXデザイン編
- ・業務アプリ開発編
- ・システム基盤編
- ・運用・保守編
- ・業務アプリ改造編
- ・(旧)再構築開発編
- SDEMテクニカルコンテンツ
- SDEMをお客様に紹介したい
- SDEM WBS選択

支援
- 研修会／説明会
- お問い合わせ
- FAQ

Pick up

* English versions of the following content are also available:
- *SDEM Standard Processes: Standard Process Structure*
- *SDEM Practical Standards: Operation and Maintenance [ITSMOP]*

SDEM continues to evolve for knowledge sharing.
In order to pass on knowledge in the field, please give us your feedback on points requiring more specifics for practical use.

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Lastly, the SDEM content introduced in this training course and the textbooks for this training can be obtained from the IKB-SDEM site.

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In order to pass on knowledge in the field, please share your experience and knowledge by giving us your feedback on points requiring more specifics for practical use.

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